

Evaluating the prediction of the toxicity of plastic chemicals

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1 Introduction

2 Dataset and problem setting

[plastik ist böse, hinführung zu degradation und additives sind böse, polyurethan domaine recycling ist schwer (chemisch)] Plastic production and consumption has a growing impact on the environment. In the past (Walker 2025)

The dataset was released by Monclús et al. 2025 in 2024 and contains 16.325 known plastic chemicals.

3 XGBoost

4 Search Strategy

5 Results

A Additional Figures

References

- Monclús, L. et al. (July 2025). “Mapping the chemical complexity of plastics”. en. In: *Nature* 643.8071, pp. 349–355. ISSN: 1476-4687. DOI: 10.1038/s41586-025-09184-8. URL: <https://www.nature.com/articles/s41586-025-09184-8> (visited on 06/30/2026).
- Walker, Tony R. (Sept. 2025). “Recycling alone cannot end the plastic pollution crisis”. In: *Cell Reports Sustainability* 2.9, p. 100521. ISSN: 2949-7906. DOI: 10.1016/j.crsus.2025.100521. URL: <https://www.sciencedirect.com/science/article/pii/S2949790625002174> (visited on 06/30/2026).